10 From Digital Heritage to Connected Data: Automating Geographical Metadata and Bibliographic Knowledge Graphs in Archeologia e Calcolatori

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Archeologia e Calcolatori, a diamond open access journal active since 1990, represents a privileged observatory of the evolution of computing in archaeology and provides today a unique corpus of knowledge on digital archaeology. Its collection of more than 1,500 articles with their related bibliographic and visual contents have been progressively transformed into an open, structured, and interoperable resource through standardized metadata, persistent identifiers, and FLOSS solutions. A significant development was implemented within the PNRR Humanities and Cultural Heritage Italian Open Science Cloud (H2IOSC) project (2022–2026), where the journal's digital heritage has been used to experiment with new approaches to data enrichment and interconnection.

This contribution presents two processes developed on this corpus. The first concerns the automated enrichment of geographical metadata through a Python-based pipeline combining toponym extraction with open-source Large Language Models (LLMs) running locally. The pipeline identifies and classifies geographical entities in texts and resolves them against gazetteers such as Pleiades and GeoNames, associating them with persistent identifiers and structured data in a FAIR-oriented approach.

The second concerns the structuring of the journal's bibliography through a workflow based on Zotero and Omeka S. Bibliographic references are transformed into structured data using shared metadata vocabularies and ontologies, including Dublin Core and BIBO, together with persistent identifiers. These data feed a knowledge graph in which publications, authors, and references are represented as interconnected entities, enabling new forms of exploration and analysis.

Together, these processes show how an established editorial heritage can be transformed through FLOSS tools into an open, interoperable, and interconnected data ecosystem, supporting the reuse of archaeological scholarship.